

Assignment 7

Sequential Agentic Workflows using LangGraph

USE CASE 1: Telecom SIM Activation & Fraud Verification Workflow

Objective

The objective of this assignment is to design and implement a Sequential Agentic Workflow using:

- LangGraph
- Google Gemini 2.5 Flash
- Tool Calling
- State Management
- Sequential Workflow Execution
- LLM-based Decision Making

This workflow automates the telecom SIM activation process by performing:

1. Customer KYC Verification
2. Fraud Detection Analysis
3. Final Activation Decision

Each stage executes sequentially, where the output of one node becomes the input for the next node.

Workflow Architecture

START -> Customer KYC Verification -> Fraud Detection Analysis ->

Final Activation Decision -> END

Workflow Explanation

Step 1 — Customer KYC Verification

Purpose

This node validates customer KYC documents such as:

- Aadhaar
- PAN

Tool Used : A custom langchain tool called **kyc_verification_tool**

The tool checks:

- Whether Aadhaar is valid
- Whether PAN is valid
- Whether any document is pending verification

This stage ensures only verified customers proceed to fraud analysis.

Step 2 — Fraud Detection Analysis using Gemini

Purpose

This node performs intelligent fraud analysis using Google Gemini 2.5 Flash.

Inputs Analyzed

Gemini analyzes:

- Customer location
- Number of previous SIM requests
- KYC verification status

Fraud Classification

Gemini classifies the customer into one of the following categories:

Classification

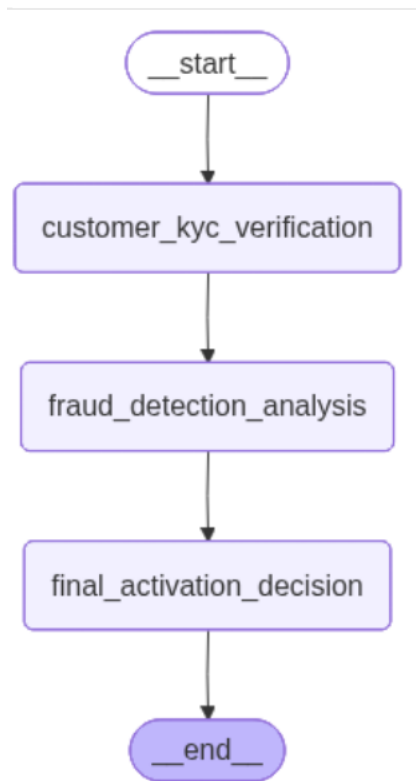
LOW_RISK	Safe customer
MEDIUM_RISK	Requires manual review
HIGH_RISK	Potential fraud

Step 3 — Final Activation Decision

Purpose

This node converts fraud classification into a business decision.

Decision Logic



USE CASE 2: Healthcare Appointment Prioritization Workflow

START -> Symptom Severity Analysis -> Medical Prioritization Analysis -> Final Consultation Assignment -> END

Workflow Explanation

Step 1 — Symptom Severity Analysis

This node evaluates the patient's medical condition based on vital symptoms.

Tool Used : symptom_severity_tool

Symptoms Analyzed

- Fever
- Oxygen Level
- Heart Rate
- Symptom Duration

Possible Outputs

Condition	Output
Normal vitals	STABLE
Moderate symptoms	MODERATE
Severe symptoms	CRITICAL

Step 2 — Medical Prioritization Analysis using Gemini

Purpose

Gemini evaluates the patient's medical urgency.

Inputs Analyzed

- Severity level
- Patient age
- Existing medical conditions

Priority Classifications

Classification

EMERGENCY	Immediate treatment required
PRIORITY_CONSULTATION	Fast specialist consultation
REGULAR_CONSULTATION	Normal consultation

Step 3 — Final Consultation Assignment

Purpose

Convert medical priority into consultation allocation.

Decision Logic

Priority	Assignment
EMERGENCY	ICU/ER
PRIORITY_CONSULTATION	Specialist Doctor
REGULAR_CONSULTATION	General Physician

